

Reflection Template

Use Case	UC-8
Use Case Name	Update Task
Requirement IDs	FR9, FR15, FR19, FR20, FR22, FR23
Matching Feature Comparison Sets	CS-18, CS-19
Implementation Order	Kotlin first -> Java second

Feature Relevance

Feature/s:

Data classes/records, null safety

Did the feature/s occur naturally in this/these context/s?

Yes, update request has nullable fields which made null safety visible again. The direct property assignment in Kotlin came up naturally when updating the task fields, so no setters were needed.

Qualitative Ratings (1-5)

- 1 = very poor/very difficult
- 2 = rather poor/difficult
- 3 = neutral/moderate
- 4 = good/easy
- 5 = very good/very easy

Criteria	Java	Kotlin
Readability	4	5
Compactness of logic	4	5
Perceived implementation effort	5	5

Justifications

Readability:

Both are readable but Kotlin's direct property assignment "task.title = request.title" reads more naturally than Java's setter calls "task.setTitle(request.title())". The nullable fields in the request are also more clearly expressed in Kotlin.

Compactness of logic:

The service logic is nearly identical in length but Kotlin's property assignment is slightly more concise than Java's setter syntax. No major difference otherwise.

Implementation effort:

Both were straightforward.

Observed structural differences

Kotlin uses direct property assignment `task.title = request.title`. Java requires setter methods `task.setTitle(request.title())`. This is something that will repeat in every update operation.

Nullable fields in `UpdateTaskRequest` are explicit in Kotlin like `String?`, `LocalDate?` but invisible in Java where the same fields are just regular types.

Both request DTOs are single-line records/data classes, no boilerplate difference here.

Final comparative summary

This UC introduced no huge new differences but reinforced existing ones. The setter vs. direct assignment difference is small but consistent and will probs repeat in future update UCs. Null safety remains more expressive in Kotlin for optional fields.